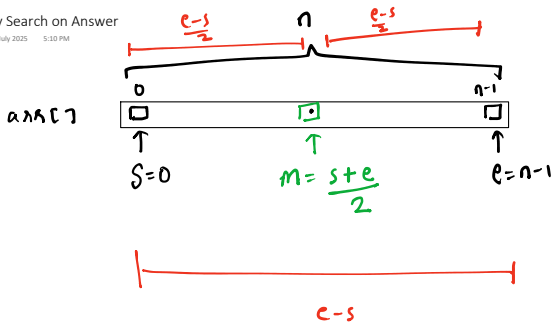




t

monotonic fn

f(x)

monotonically
↑ing fn

f(x)

monotonically
↓ing fn

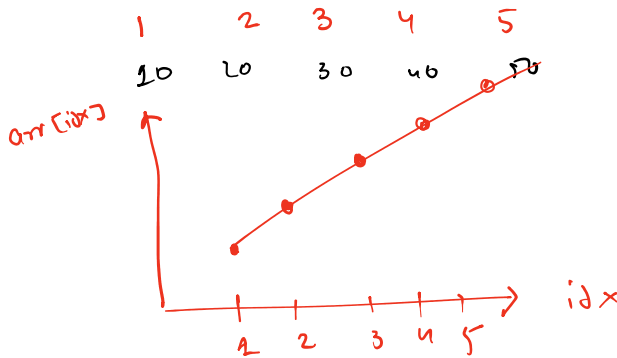
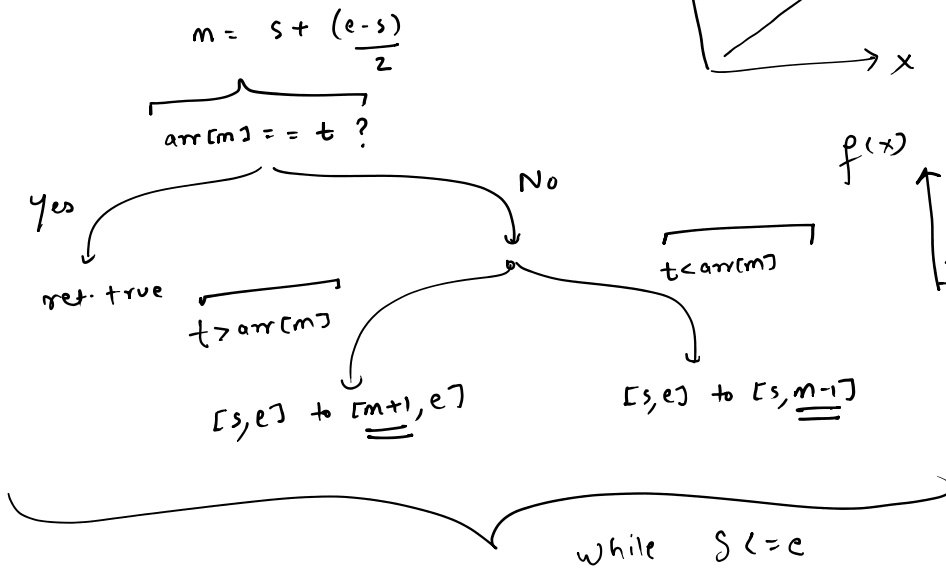
f(x)

f(x)

monotonically
non-dec.

f(x)

monotonically
non-inc



Binary Search on Answer

[6.5]

[10k hours]
1440

ans ∈ [0, 10,000 hr]

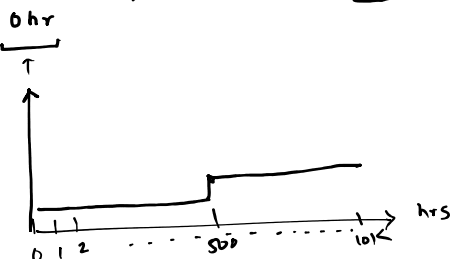
1 day = 4 hr

10 day = 40 hr

30 day = 120 hr

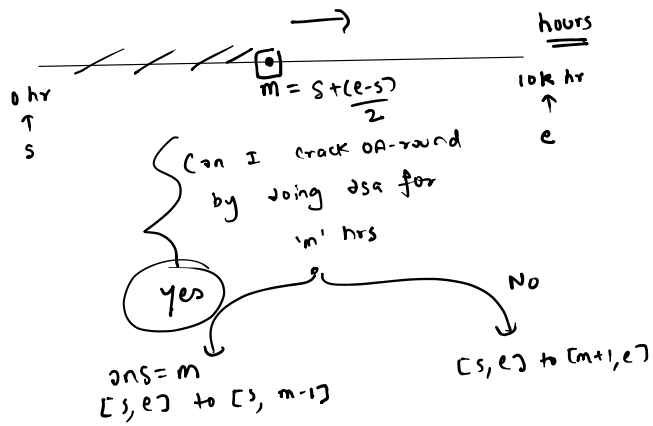
track a-round? 1y = 120 × 12 = 1440 hr

No	No	No	...	No	...	No	Yes	Yes	Yes	...	Yes
1	1	1	...	1	...	1	1	1	1	...	1
0 hr	1	2	...	100 hr	...	500 hr	500 hr	500 hr	500 hr	...	10,000 hr

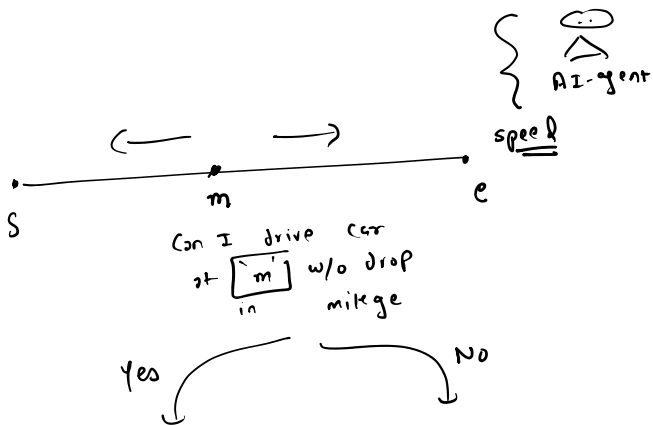
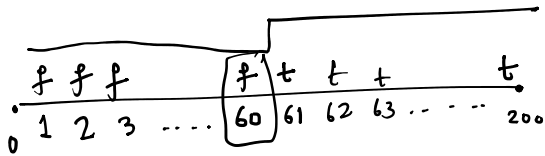


predicate fn

f(x) { AI agent }



find the max. speed at which I can drive w/o drop in mileage?

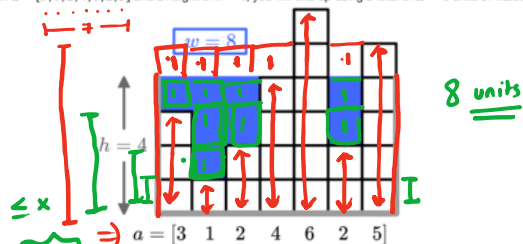


Building Aquarium

You love fish, that's why you have decided to build an aquarium. You have a piece of coral made of n columns, the i -th of which is a_i units tall. Afterwards, you will build a tank around the coral as follows:

- Pick an integer $h \geq 1$ — the height of the tank. Build walls of height h on either side of the tank.
- Then, fill the tank up with water so that the height of each column is h , unless the coral is taller than h ; then no water should be added to this column.

For example, with $a = [3, 1, 2, 4, 6, 2, 5]$ and a height of $h = 4$, you will end up using a total of $w = 8$ units of water, as shown.



You can use at most x units of water to fill up the tank, but you want to build the biggest tank possible. What is the largest value of h you can select?

$h=1$ ✓ $h=2$ ✓ $h=3$ ✓ $h=4$ ✓ $h=5$ ✗ $h=6$ ✗ $h=7$ ✗ ...

Input

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

The first line of each test case contains two positive integers n and x ($1 \leq n \leq 2 \cdot 10^5$; $1 \leq x \leq 10^9$) — the number of columns of the coral and the maximum amount of water you can use.

The second line of each test case contains n space-separated integers a_i ($1 \leq a_i \leq 10^9$) — the heights of the coral.

The sum of n over all test cases doesn't exceed $2 \cdot 10^5$.

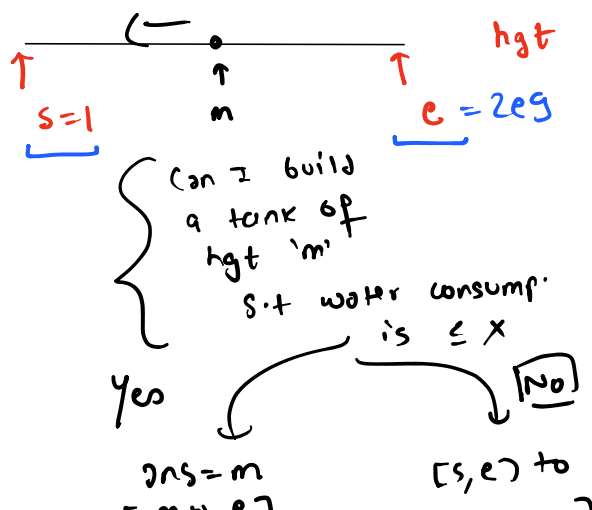
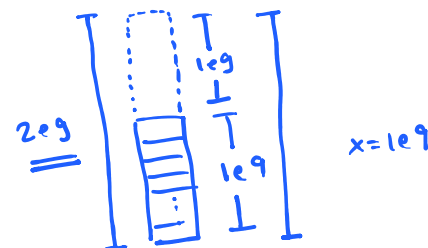
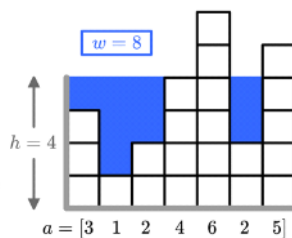
Output

For each test case, output a single positive integer h ($h \geq 1$) — the maximum height the tank can have, so you need at most x units of water to fill up the tank.

We have a proof that under these constraints, such a value of h always exists.

Example

input	Copy
5	
3 1 2 4 6 2 5	
3 10	
1 1 1	
4 1	
1 4 3 4	
6 1984	
2 6 5 9 1 8	
1 1000000000	
1	
output	Copy
4	
4	
2	
335	
1000000001	



' ↓
ans = m
[m+1, e]

u
[s, e] to
[s, m-1]

Minimum days to make m bouquets

You are given an integer array **bloomDay**, an integer **m** and an integer **k**. You want to make **m** bouquets. To make a bouquet, you need to use **k adjacent flowers** from the garden. The garden consists of **n** flowers, the i^{th} flower will bloom in **bloomDay[i]** days and then can be used in **exactly one** bouquet. Return the **minimum number of days** you need to wait to be able to make **m bouquets** from the garden. If it is impossible to make m bouquets return -1.

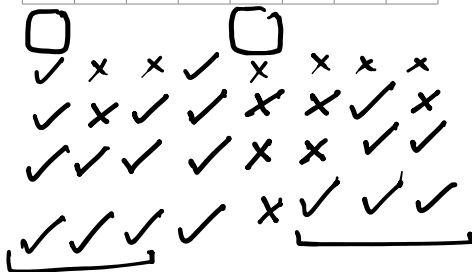
Constraints

$1 \leq n \leq 10^5$
 $1 \leq \text{bloomDay}[i] \leq 10^9$
 $1 \leq m \leq 10^6$
 $1 \leq k \leq n$

Input

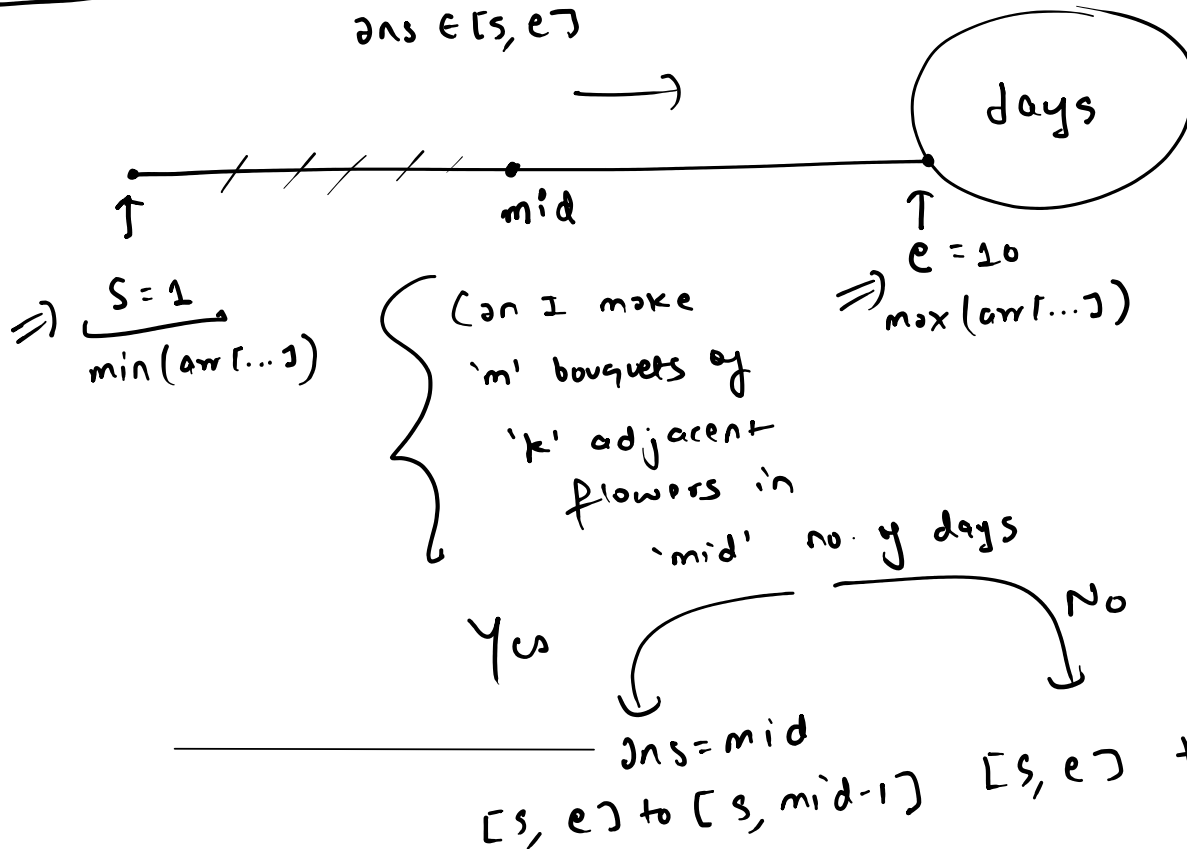
$n = 8$ $m = 2$ $k = 3$

0	1	2	3	4	5	6	7
1	3	2	1	10	4	2	3



ans $\in [s, e]$

No No No Yes Yes Yes ... Yes
 day-1 day-2 day-3 day-4 day-5 day-6 ...



Hamburgers



Polycarpus loves hamburgers very much. He especially adores the hamburgers he makes with his own hands. Polycarpus thinks that there are only three decent ingredients to make hamburgers from: a bread, sausage and cheese. He writes down the recipe of his favorite "Le Hamburger de Polycarpus" as a string of letters 'B' (bread), 'S' (sausage) и 'C' (cheese). The ingredients in the recipe go from bottom to top, for example, recipe "BSCBS" represents the hamburger where the ingredients go from bottom to top as bread, sausage, cheese, bread and sausage again.

Polycarpus has n_b pieces of bread, n_s pieces of sausage and n_c pieces of cheese in the kitchen. Besides, the shop nearby has all three ingredients, the prices are p_b rubles for a piece of bread, p_s for a piece of sausage and p_c for a piece of cheese.

Polycarpus has r rubles and he is ready to shop on them. What maximum number of hamburgers can he cook? You can assume that Polycarpus cannot break or slice any of the pieces of bread, sausage or cheese. Besides, the shop has an unlimited number of pieces of each ingredient.

Input

The first line of the input contains a non-empty string that describes the recipe of "Le Hamburger de Polycarpus". The length of the string doesn't exceed 100, the string contains only letters 'B' (uppercase English B), 'S' (uppercase English S) and 'C' (uppercase English C).

The second line contains three integers n_b, n_s, n_c ($1 \leq n_b, n_s, n_c \leq 100$) — the number of the pieces of bread, sausage and cheese on Polycarpus' kitchen. The third line contains three integers p_b, p_s, p_c ($1 \leq p_b, p_s, p_c \leq 100$) — the price of one piece of bread, sausage and cheese in the shop. Finally, the fourth line contains integer r ($1 \leq r \leq 10^{12}$) — the number of rubles Polycarpus has.

Please, do not write the `%lld` specifier to read or write 64-bit integers in C++. It is preferred to use the `cin`, `cout` streams or the `%I64d` specifier.

Output

Print the maximum number of hamburgers Polycarpus can make. If he can't make any hamburger, print 0.

Examples

input	Copy
BBBSSC 6 4 1 1 2 3 4	
output	Copy
2	

Homework

<https://leetcode.com/problems/find-the-smallest-divisor-given-a-threshold/>

<https://leetcode.com/problems/koko-eating-bananas/>

<https://leetcode.com/problems/search-a-2d-matrix/>

Painters Partition

Given **K** painters to paint **N** boards where each painter takes one unit of time to paint one unit of a board. Design an algorithm to compute the **minimum amount of time** required by the **K** painters to paint all the **N** boards – such that a painter can only paint **contiguous segments of boards** and a board can only be painted, at maximum, by a **single painter**.

Constraints

- $1 \leq K \leq 10$
- $1 \leq N \leq 10$
- $1 \leq \text{length of each board} \leq 10^8$

Example

- Input : $K=3, N=5$ [1, 3, 5, 2, 4]
- Output : 6

1	3	5	2	4
---	---	---	---	---

_____ **time**

_____ **time**

1	3	5	2	4
----------	----------	----------	----------	----------

_____ time
5 10

Square Root of a Number

Given a non-negative integer N , design an algorithm to find its **square root**.

Example

Input : $N = 64$
Output : 8

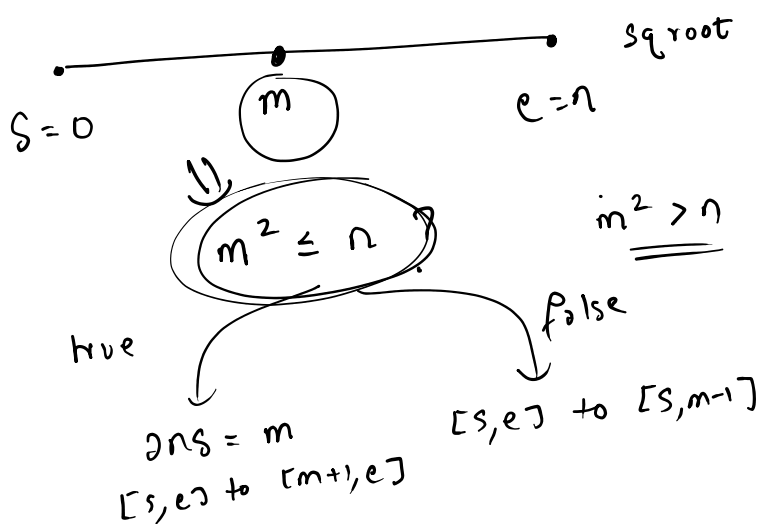
Input : $N = 30$
Output : 5.47

$$7 = 49$$

$$\Rightarrow 1. \sqrt{n} \geq 0$$

$$2. (\sqrt{n})^2 \leq n$$

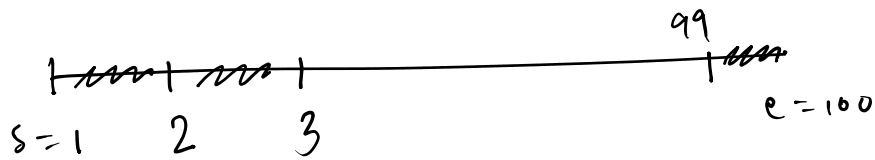
$$\begin{array}{ccccccc} 0^2 & 1^2 & 2^2 & \dots & 7^2 & 8^2 & 9^2 & 10^2 & \dots \\ \checkmark & \checkmark & \checkmark & & \checkmark & \times & \times & \times & \dots \times \end{array}$$



$$\begin{array}{l} 5 \cdot 0^2 \leq 30? \checkmark \\ 5 \cdot 1^2 \leq 30? \checkmark \\ 5 \cdot 2^2 \leq 30? \checkmark \\ 5 \cdot 3^2 \leq 30? \checkmark \\ 5 \cdot 4^2 \leq 30? \checkmark \\ 5 \cdot 5^2 \leq 30? \times \end{array}$$

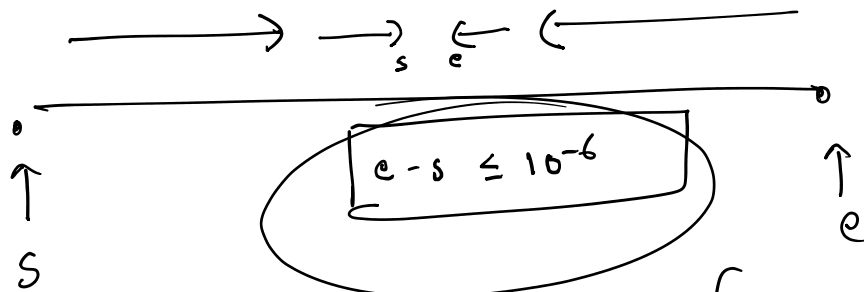
$$\begin{array}{l} 5.41^2 \leq 30? \checkmark \\ 5.42^2 \leq 30? \checkmark \\ \vdots \\ 5.47^2 \leq 30? \checkmark \\ 5.48^2 \leq 30? \times \end{array}$$

$$\begin{array}{l} s=1 \\ e=100 \\ \log_{100} \sim 7 \text{ iterations} \end{array}$$



$$\text{error} = 10^{-6}$$

$$\text{your o/p} = \text{actual o/p} \pm \text{error}$$

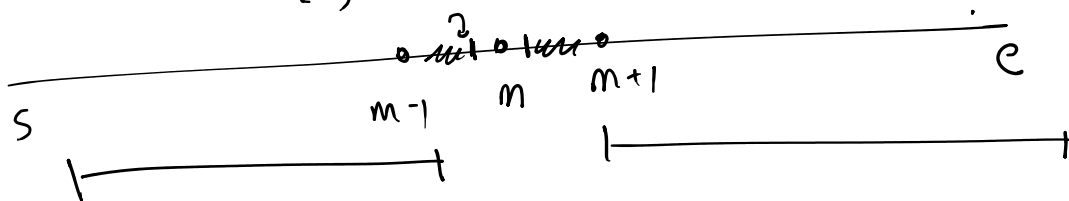


$$\text{while } s \leq e \left\{ \begin{array}{l} \text{while } e-s > 10^{-6} \end{array} \right. \text{error} = 10^{-6}$$

}

$$[s, e] \rightarrow [m+1, e]$$

$$[s, e] \rightarrow [s, m-1]$$



$$\left\{ \begin{array}{l} [s, e] \rightarrow [m + \text{error}, e] \\ [s, e] \rightarrow [s, m - \text{error}] \end{array} \right.$$

$$\text{error} = 10^{-16}$$

$$\frac{e-s}{2^0} = e-s > 10^{-16}$$

$$= e-s$$

$$\frac{e-s}{2^1}$$

$$\frac{1}{2}$$

$$\frac{e-s}{2^2} = \frac{e-s}{4}$$

$$\frac{e-s}{8}$$

$$\left(\frac{e-s}{2^K} \right) \leq 10^{-16}$$

$$2^K \geq \frac{e-s}{10^{-16}}$$

$$s=0$$

$$e=n$$

$$e-s = n-0 = n$$

$$n \leq 10^6$$

$$2^K \geq (e-s) 10^{16}$$

$$2^K \geq 10^6 \cdot 10^{16}$$

$$2^K \geq 10^{22}$$

$$K = \log_2(10^{22}) \sim 75$$

$$2^{10} = 1024 \sim 10^3$$

$$2^{20} \sim 10^6$$

$$2^{30} \sim 10^9$$

⋮

$$2^{60} \sim 10^{18}$$

$$2^{70} \sim 10^{21}$$

$$2^{80} \sim 10^{24}$$

⋮

Ropes

There are n ropes, you need to cut k pieces of the same length from them. Find the maximum length of pieces you can get.

Input

The first line contains two integers n and k ($1 \leq n, k \leq 10000$). Next n lines contain one number each, the length of the rope a_i ($1 \leq a_i \leq 10^7$).

Output

Output one real number, maximum length of pieces you can get. The answer will be considered correct if the relative or absolute error does not exceed 10^{-6} .

Example

input	Copy
4 11 802 743 457 539	
output	Copy
200.5	

Aggressive Cows

A Farmer has built a **new long barn**, with **N** ($2 \leq N \leq 10^5$) **stalls**. The stalls are located along a **straight line** at positions $x_1 \dots x_N$ ($0 \leq x_i \leq 10^9$). His **C** ($2 \leq C \leq N$) **cows** don't like this barn layout and become **aggressive** towards each other once put into a stall.

To prevent the cows from hurting each other, the farmer wants to **assign the cows** to the stalls, such that the **minimum distance** between any **two of them** is as **large** as possible.

What is the **largest minimum distance**?

Example

Input : C=3, N=5 [1, 2, 4, 8, 9]

Output : 3

Roti Prata

IEEE is having its AGM next week and the president wants to serve cheese prata after the meeting. The sub-committee members are asked to go to their food connections and get **P** ($P \leq 1000$) **pratas** packed for the function. The stall has **L** ($L \leq 50$) **cooks** and each cook has a **rank R** ($1 \leq R \leq 8$). A cook with a rank **R** can cook one prata in the first **R** minutes, one more prata in the next **2R** minutes, one more prata in the next **3R** minutes and so on (he can only cook a complete prata) - for instance, if the rank of a cook is 2 then he will cook one prata in first 2 minutes, one more prata in the next 4 mins and one more in the next 6 minutes hence in total 12 minutes he cooks 3 pratas, in 13 minutes also he can cook only 3 pratas as he does not have enough time for the 4th prata. Design an algorithm to compute the **minimum time** to get the order done.

Example

- **Input : P=10, L=4 [1, 2, 3, 4]**
- **Output : 12**

Median of Two Sorted Arrays

https://youtu.be/dvZEMMB_jis?t=403

Given two sorted arrays, **nums1** & **nums2**, containing **M** & **N** integers respectively design an algorithm to find the **median** of the two arrays.

Example

Input

1	3	5	7		
0	2	4	6	8	9

Output : 4.5

Input

1	3	5	7	
0	2	4	6	8

Output : 4